

Contact-State Recovery for Cable-Connector Insertion: Paired Force-Safety Tests and a Direct MuJoCo Audit

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DRAFT—NOT FOR SUBMISSION

Abstract—Cable-connector insertion couples visual misalignment, cable load, frictional contact, and damage-sensitive parts. We present ContactBelief, a contact-state recovery controller, and a paired protocol separating algorithmic from robot-execution evidence. The 2.5-D benchmark evaluates eight policies on 600 shared scenarios (4800 episodes). In the hard condition, ContactBelief achieves 90.0% verified, undamaged success, versus 56.5% for direct insertion and 46.5% for spiral search. The paired gains of 33.5 and 43.5 points remain significant after Holm correction. Removing orientation updates lowers success to 78.5%; removing retract lowers it to 84.0%. Guarded admittance reaches 86.0% success with lower mean peak force, and ContactBelief does not significantly outperform it. A 300-episode direct MuJoCo audit uses current observations and native contact termination: ContactBelief reaches 60.0% success and 0.0% damage, versus 25.0% and 75.0% for direct insertion. These results support contact-state recovery under two synthetic physics abstractions, not real connector reliability, torque-level control, hardware safety, or factory readiness.

Note to Practitioners— Small terminal-pose errors, cable drag and friction can turn automated connector insertion into a jam or a damaged part. This study evaluates a recovery sequence that detects unexpected contact, unloads the terminal, updates alignment and then verifies locking. In the synthetic benchmark it is substantially safer than blind direct insertion and spiral search, but it does not significantly outperform a carefully tuned guarded-admittance controller. The practical message is therefore not to replace a reliable force-control baseline with an unverified learned controller. A deployment should first measure connector clearance, insertion force, latch behavior and cable reaction for each product family, then compare controllers on paired fixtures while recording peak force, damage, cycle time, retries and a post-insertion pull test. The MuJoCo robot in this paper is a task-space digital twin, not evidence of production reliability. The proposed hardware gate specifies the measurements required before factory claims are made.

Index Terms—Contact-rich manipulation, force control, industrial automation, robotic assembly, wire-harness manufacturing.

I. Introduction

Insertion is often presented as a rigid peg-in-hole problem, yet an electrical connector at the end of a wire harness

is not simply a peg. Upstream cable bending creates a load that changes with approach depth; the grasped terminal can rotate; socket chamfers and latches create discontinuous contact; and a successful-looking insertion must still be locked and verified. Classical compliance and force control provide the right low-level language [14], [4], but the high-level question remains: when should a robot keep inserting, when should it infer edge contact, and which corrective state should follow?

Learning-based insertion has demonstrated variable compliance, force-only policies, visual servoing, residual control, and simulation-to-real transfer [1], [9], [3], [7], [12]. Recent work also targets unseen industrial connectors and deformable cable insertion [11], [16]. Those systems motivate hardware experiments that this paper does not yet possess. Our immediate goal is more modest and falsifiable: construct an auditable synthetic benchmark in which the same latent physical scenario is presented to each controller, define damage and lock verification before the formal run, retain strong classical and oracle comparators, and test whether the controller behavior survives a second physics abstraction with native contacts.

ContactBelief is a finite-state engineering controller, not a learned policy and not a Bayesian posterior. Its “belief” is the controller’s internal discrete estimate of free-space approach, guarded insertion, edge contact, retraction, realignment, and verified seating. Contact wrench signs update the visual alignment target; a retract state unloads the guide before the next probe. The paper’s contribution is the controller/evidence package rather than a claim that this state machine supersedes modern learning systems.

The contributions are:

- 1) a paired 2.5-D connector-insertion benchmark with clearance, friction, cable drag, visual and force bias, contact wrench, jamming, damage, retry, and six-step lock verification;
- 2) eight frozen policies, three difficulties, 200 common-random-number scenarios per difficulty, exact paired tests, bootstrap intervals, and orientation/retract ablations;
- 3) a collision-enabled terminal and socket in MuJoCo [13], with controllers observing native state and contact force directly over 300 additional paired episodes; and

- 4) an episode-disjoint public-data ingestion audit plus a 153-check independent evidence audit that prevents offline or synthetic metrics from being reported as hardware performance.

II. Related Work

a) Compliance and force-guided insertion.: Remote-center compliance and impedance control established that suitable mechanical or commanded compliance can convert contact into alignment rather than wedging [14], [4]. High-level search, force regulation, and insertion remain sensitive to contact stiffness. Lee and Pham demonstrate one robust controller across a 10–100 N/mm stiffness range, illustrating why a strong force baseline is necessary [8]. ContactBelief is evaluated against guarded admittance and does not outperform it significantly in our hardest primary condition.

b) Learning assembly behavior.: Variable-compliance reinforcement learning can combine a nominal trajectory, force response, and learned residual action [1]; residual RL similarly retains an engineered controller while learning corrections [7]. Force-only offline deep Q-learning has reported tight-clearance insertion on hardware [9]. Continuous visual servoing can reduce the uncertainty region before compliant contact and avoid blind search [3]. Our baselines isolate these ingredients—vision without force, spiral search, guarded force recovery, and two state-machine ablations—without claiming to reproduce their full systems.

c) Connectors, deformable cables, and simulation.: Industrial insertion policies can be adapted to unseen connectors by online practice [11]. Wu et al. explicitly address a cable-held Type-C connector and exploit contact during curriculum RL [16]. IndustReal shows substantially stronger evidence than ours: contact-rich simulation, policy transfer, and a real assembly system [12]. Simulation nevertheless has a reality gap, especially for stiff contact. We therefore keep the 2.5-D and MuJoCo results separate, never pool them, and reserve hardware claims for future experiments.

III. Contact-State Method

A. State and sensing

The primary state is $s_t = (x_t, y_t, \psi_t, \theta_t)$: axial insertion depth, lateral displacement, yaw, and pitch. A 30-mm terminal maps orientation to a tip offset

$$e_t = y_t + 0.030\psi_t + 0.45(0.030)\theta_t, \quad (1)$$

and the collision envelope is

$$E_t = |e_t| + 0.38(0.030)(|\psi_t| + |\theta_t|). \quad (2)$$

The controller observes biased/noisy pose $\hat{s}_t$, axial force F_x , lateral force F_y , and contact torque τ_c . Bias is fixed within an episode; noise is a frozen time sequence shared by all policies with the same seed. This pairing prevents one controller from receiving an easier draw.

B. Contact update and recovery

During free-space approach, visual servoing moves the observed pose toward an internal target $(\bar{y}, \psi, \theta)$. An edge-contact event is declared when lateral force exceeds 0.9 N or axial force approaches the scenario’s jam threshold. Contact load is

$$L_t = \max\{|F_y|, F_x - 5\}, \quad (3)$$

and the target shifts opposite the lateral contact sign,

$$\bar{y} \leftarrow \bar{y} - \text{sgn}(F_y) \text{clip}(0.00055L_t, 0.0010, 0.0042). \quad (4)$$

Torque sign also shifts yaw and pitch targets. The controller retracts for $8 + \lfloor 3d \rfloor$ steps at difficulty d , realigns in unloaded space, and then resumes a slower probe. The guarded-admittance comparator uses a higher force threshold, lateral-only correction, and a shorter retract. The no-orientation ablation preserves contact and retract but omits angular target updates. The no-retract ablation corrects while loaded; its proxy increases effective contact load by 1.45 and accumulates damage impulse above 8 N rather than 10 N, representing sliding friction and edge pressure.

C. Contact, jam, damage, and success

For clearance c , violation is $v_t = \max(0, E_t - c)$. Near the socket, contact force is a stiffness-scaled function of v_t and penetration, added to a friction, drag, and insertion-speed seating load. If total load exceeds the scenario jam threshold, axial motion is attenuated until retraction unloads the contact. Damage occurs when either peak force exceeds a scenario limit or the integral of high force exceeds a scenario impulse limit. Success requires 40 mm depth, $E_t \leq 0.9c$, no damage, and six consecutive lock-verification steps. An episode ends at verified success, damage, or 320 steps (6.4 s). These equations are engineering proxies, not a finite-element connector model.

IV. Paired Evaluation Protocol

A. Scenario distribution

Difficulty changes uncertainty rather than policy parameters. Clearance is sampled from $[7.5, 10.5](1 - 0.43d)$ mm. Friction is $U(0.14, 0.28) + dU(0.20, 0.42)$; contact stiffness is 1.5–2.1 kN/m multiplied by $1 + 0.35d$. Initial lateral error spans $\pm(13 + 15d)$ mm; yaw spans $\pm(0.10 + 0.20d)$ rad and pitch 80% of that range. Visual-bias and noise scales grow with d , as do three cable-drag components. Jam, peak-force, and impulse limits vary per scenario. The exact equations, seed schedule, and pilot history were frozen in a protocol before the formal run.

For each $d \in \{0.2, 0.5, 0.8\}$, 200 latent scenarios are constructed from base seed 83,000,000. Eight policies receive every scenario: ContactBelief, guarded admittance, vision-staged without force, spiral search, direct insertion, the two ablations, and a state-aware oracle. The oracle reads latent pose and is an upper reference, not a deployable comparator. There are no exclusions.

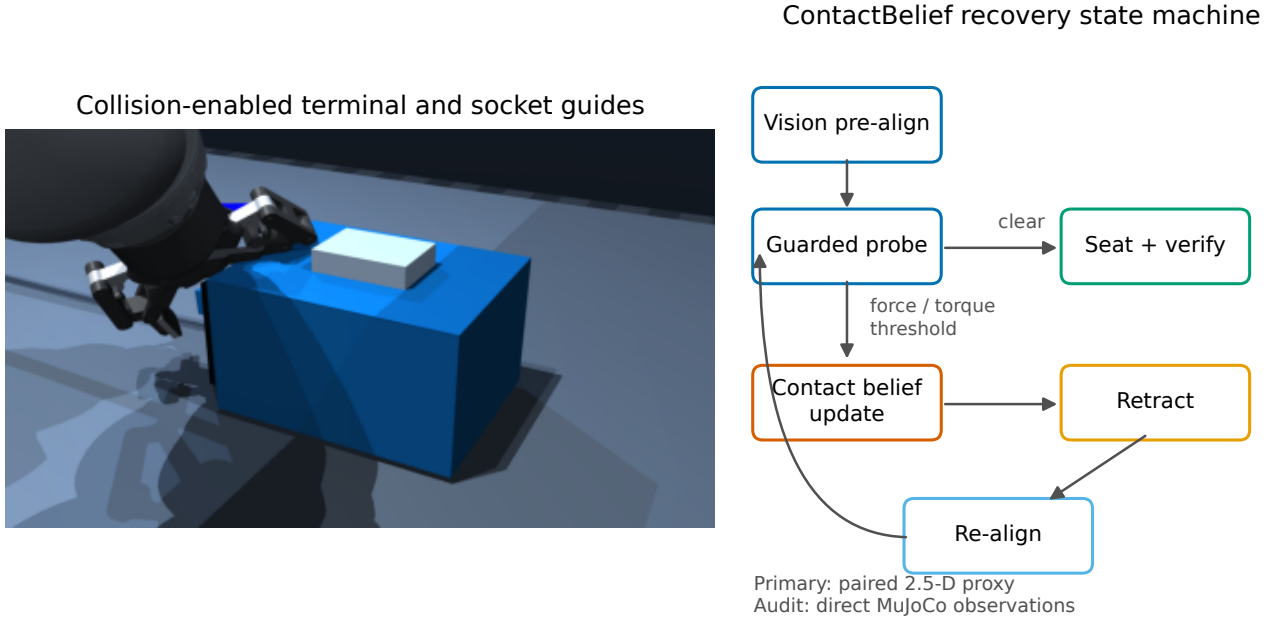


Fig. 1. InsertBot evidence and controller overview. Left: the realistic KUKA/Robotiq visual twin surrounds a compliant, collision-enabled terminal and four-rail socket in the direct MuJoCo audit. Right: ContactBelief updates its target from contact signs, unloads the guide, realigns, and then retries. Primary results use a paired 2.5-D proxy; no trajectory or contact event is replayed into the direct MuJoCo controller.

B. Endpoints and statistics

The primary endpoint is verified, undamaged success. Secondary endpoints are first-pass success, damage, jam occurrence, peak force, cycle time, retries, and failure type. Per-policy success and mean force receive deterministic 10,000-draw nonparametric bootstrap intervals [2]. Within each difficulty, ContactBelief is compared with seven policies using exact paired McNemar tests [10]; paired force uses a one-sided Wilcoxon signed-rank test [15]. Holm correction is applied separately across all 21 success tests and all 21 force tests [5]. Effect sizes and intervals are reported even when tests are not significant.

C. Public-data auxiliary audit

We retain the MIT-licensed LeRobot ALOHA simulated-insertion dataset as an explicitly mismatched public proxy [6]. Its hashed cache contains 25000 finite state-action frames from 50 complete 500-frame episodes. A seeded whole-episode split uses 40 episodes for fitting and 10 for test, with no shared episode. A 30-input multi-output ridge residual model reaches held-out action RMSE 0.03271, versus 0.13021 for $a_t = s_t$ (74.88% lower). This validates data ingestion, splitting, and an offline action interface only. The rigid dual-arm simulation has no automotive wire terminal, cable reaction, camera observation, or contact force; it is excluded from every success and force claim below.

V. Primary Results

A. Difficulty exposes the contact gap

Easy scenarios are nearly saturated: ContactBelief and the oracle reach 100%, while direct and visual-only insertion reach 99.5%. At medium difficulty, ContactBelief reaches 94.5%, compared with 90.0% guarded, 83.0% spiral, and 80.5% direct/vision-only. Hard difficulty separates the methods: ContactBelief reaches 90.0% (95% CI 86.0–94.0), direct and vision-only each 56.5%, and spiral 46.5%. ContactBelief’s damage rate is 4.5%, compared with 37.5% for direct insertion and 53.5% for spiral search.

The oracle remains 100% at all difficulties and therefore shows that the task distribution itself is solvable. ContactBelief is still ten points below that ceiling in the hard condition, leaving a meaningful estimation/control gap.

B. Paired effects and ablations

At hard difficulty, ContactBelief succeeds on 69 seeds where direct insertion fails; direct succeeds on only two reciprocal seeds. This yields a 33.5-point effect (bootstrap CI 26.5–40.0) and $p = 4.33 \times 10^{-17}$ after correction. Mean peak force is $|-1.5|$ N lower (paired CI 1.39–1.65 N). Against spiral search the success gain is 43.5 points. Thus the main conclusion is not merely that recovery takes longer: it increases verified success while reducing the high-force burden of force-blind insertion.

Removing orientation update reduces hard success from 90.0% to 78.5% (11.5 points, Holm $p = 0.011$) and raises mean peak force by 3.8 N (Holm $p = 1.43 \times 10^{-6}$). Removing retract reaches 84.0%; the six-point decrement

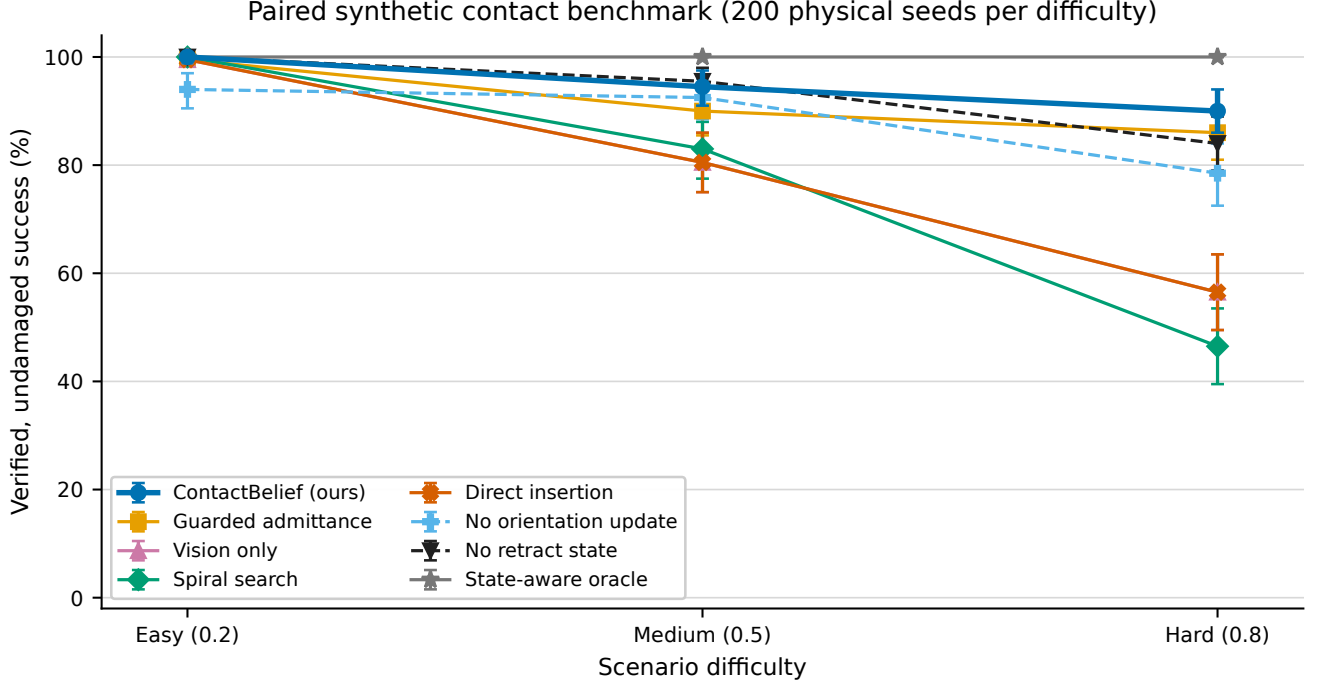


Fig. 2. Verified, undamaged success with deterministic bootstrap 95% intervals. Differences are small in the easy condition. Force-blind baselines degrade sharply as bias, drag, friction, and pose error increase. The state-aware oracle is a ceiling check.

TABLE I

Primary paired benchmark. Each policy–difficulty cell contains 200 episodes. Peak force and time are means in the hard condition.

Policy	Easy	Mid	Hard	Hard dmg.	Hard force	Hard time
		success (%)		(%)	(N)	(s)
ContactBelief (ours)	100.0	94.5	90.0	4.5	10.4	2.92
Guarded admittance	99.5	90.0	86.0	3.5	9.8	3.51
Vision only	99.5	80.5	56.5	32.5	10.9	2.55
Spiral search	100.0	83.0	46.5	53.5	15.4	2.30
Direct insertion	99.5	80.5	56.5	37.5	11.9	1.80
No orientation update	94.0	92.5	78.5	15.0	14.2	2.88
No retract state	100.0	95.5	84.0	13.5	11.7	2.56
State-aware oracle	100.0	100.0	100.0	0.0	8.1	1.58

TABLE II

Hard-condition paired comparisons, ContactBelief minus comparator. Negative Δ force favors ContactBelief. Holm correction covers 21 tests per endpoint family.

Comparator	Δ succ. (pp)	Holm p McNemar	Δ force (N)	Holm p Wilcoxon
Guarded admittance	+4.0	1.000	+0.6	1.000
Vision only	+33.5	4.33×10^{-17}	-0.4	1.49×10^{-25}
Spiral search	+43.5	6.11×10^{-24}	-5.0	1.42×10^{-25}
Direct insertion	+33.5	4.33×10^{-17}	-1.5	5.43×10^{-30}
No orientation update	+11.5	0.011	-3.8	1.43×10^{-6}
No retract state	+6.0	0.186	-1.3	0.075
State-aware oracle	-10.0	3.05×10^{-5}	+2.3	1.000

and 1.3-N force increase do not remain significant after correction ($p = 0.186$ and 0.075). We therefore claim strong primary evidence for orientation correction but only

directional primary evidence for the retract state.

Guarded admittance is the important counterexample to overclaiming. It reaches 86.0%, only four points below ContactBelief; the paired CI includes zero and Holm $p = 1$. Its mean peak force is 0.6 N lower, while mean cycle time is 0.59 s longer. The formal result supports parity-to-small-gain against this strong baseline, not superiority.

VI. Direct MuJoCo Contact Audit

A. Independent closed loop

The second backend uses a $50 \times 12 \times 12$ -mm collision-enabled terminal, four guide rails, a backstop, and a compliant weld to a commanded gripper target. Each scenario varies the 3-D start pose, opening, friction, visual/force

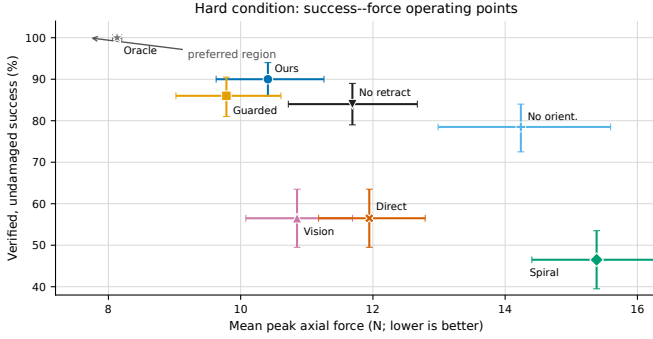


Fig. 3. Hard-condition operating points with bootstrap 95% intervals. ContactBelief occupies a favorable success-force region, but guarded admittance has slightly lower mean force and the oracle remains superior.

TABLE III

Direct MuJoCo audit at difficulty 0.8 ($n = 20$ paired scenarios per policy).

Policy	Succ. (%)	Damage (%)	Peak force (N)	Time (s)
ContactBelief (ours)	60.0	0.0	25.8	1.77
Guarded admittance	60.0	0.0	23.7	1.91
Direct insertion	25.0	75.0	102.6	0.43
No orientation update	60.0	5.0	35.4	1.74
No retract state	35.0	65.0	67.2	0.84

bias, and noise. The controller observes the current MuJoCo body/site pose and native `mj_contactForce` values at every outer-loop step. Contact-aware methods have a physics-rate force stop; direct insertion does not. The 2.5-D trajectory, contact event, force, and outcome are never replayed.

Twenty paired seeds at each difficulty and five policies give 300 episodes. A seated terminal must remain within axial, lateral, vertical, yaw, and pitch tolerances for five updates without crossing a scenario damage-force proxy. The KUKA and Robotiq meshes are visual; the audit remains task-space control. The cable visual/dynamic chain is attached to the kinematic gripper target and does not transmit its full reaction force into the compliant terminal. This is an explicit limitation.

At hard difficulty, ContactBelief and guarded admittance both reach 60.0% success with 0.0% damage. Direct insertion falls to 25.0% success with 75.0% damage; the no-retract ablation reaches 35% success with 65.0% damage. No-orientation reaches 60% success but incurs 5% damage and higher mean peak force than ContactBelief. The 35-point success difference versus direct insertion is not significant after correcting all 12 MuJoCo comparisons (Holm $p = 0.156$); it is retained as descriptive evidence. Equal success and slightly lower mean force for guarded admittance again prevent a superiority claim.

VII. Limitations and Hardware Gate

The result is a research draft, not a submission-ready hardware paper. First, the 2.5-D contact equations and

damage thresholds are engineered proxies and were calibrated through retained pilot runs. The formal seeds are disjoint from pilots, but the protocol was not registered externally. Second, ContactBelief is hand-designed; its state is not a probability distribution and it has not been trained against diverse connector families. Third, the direct MuJoCo socket is deliberately generous and simplified. It omits latch mechanics, plastic deformation, pin-level geometry, measured camera latency, force-sensor bandwidth, robot torque limits, and a cable-to-terminal reaction path. Fourth, a synthetic damage event is not evidence that a real connector would be damaged at that force. The public ALOHA audit is also a rigid simulated task-mismatch control, not real insertion evidence.

A credible hardware gate should freeze at least three connector/cable families, measure actual clearance and force limits, randomize fixture and grasp pose, calibrate camera and six-axis force/torque sensing, and preregister success, peak force, cycle time, jam, part damage, and a post-insertion pull test. The paired unit should be the same physical fixture perturbation across policies, with at least 100 trials per key comparison and no unreported exclusions. Guarded admittance must remain the primary classical comparator. Only after passing this gate should the paper make claims about factory reliability, service life, labor substitution, or payback.

VIII. Reproducibility and Audit

The primary protocol freezes distributions, seeds, hypotheses, and multiplicity handling before its 200-seed-per-difficulty run. A separate MuJoCo protocol records nine retained calibration iterations and freezes the collision fixture before the 20-seed formal audit. Episode CSVs, aggregate CSVs, paired tests, protocols, sources, scene XML, representative trace, and image are SHA-256 hashed. The public dataset manifest, nine source files, portable cache, episode split, model, and offline metrics are audited separately. An independent script also recomputes row counts, unique paired keys, seed schedules, scenario invariance, success/damage invariants, every aggregate, bootstrap interval, exact McNemar statistic, Holm correction, and paired Wilcoxon result. All 153 checks pass. Generated tables and figures are accepted only from that passing audit.

IX. Conclusion

Contact-triggered target updates, orientation correction, unloading, and retry substantially improve a cable-connector insertion proxy over force-blind direct and spiral approaches. The strongest result is paired: at hard difficulty, ContactBelief gains 33.5 points over direct insertion while lowering peak force. Orientation update is supported by corrected ablation tests; retract is directional in the primary benchmark and strongly safety-relevant in the direct MuJoCo audit. Guarded admittance matches or nearly matches ContactBelief in both backends, so it remains the honest baseline to beat. The next scientific step is not

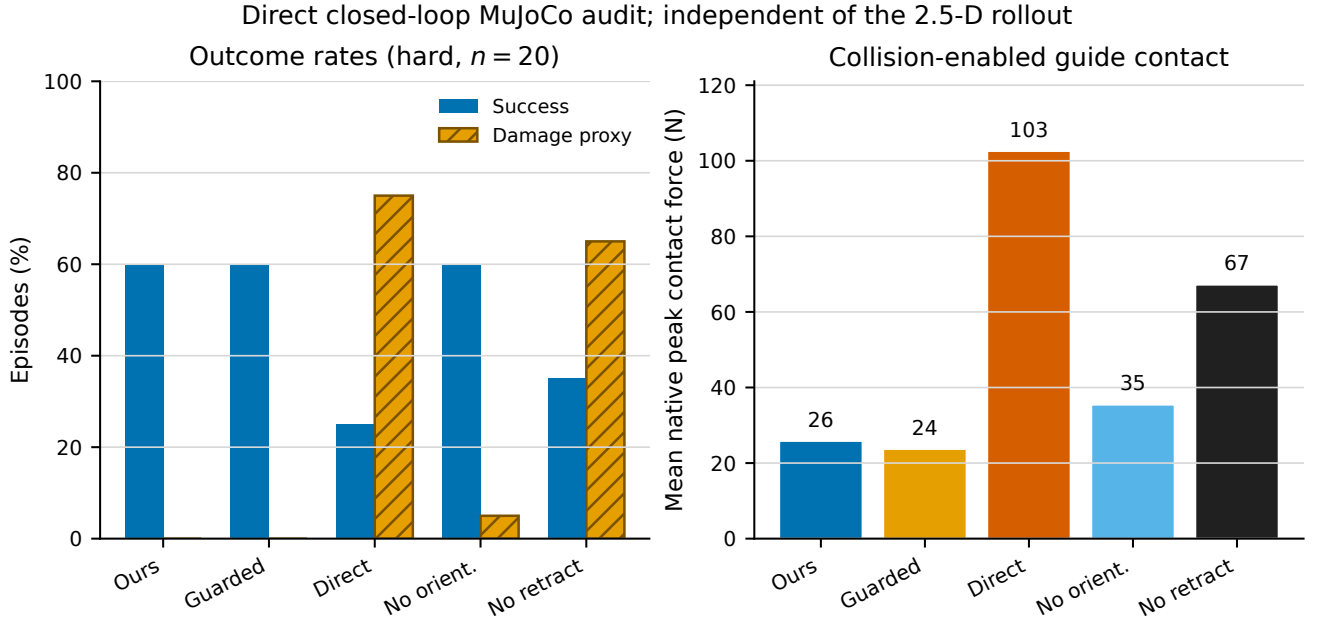


Fig. 4. Direct MuJoCo hard-condition audit. ContactBelief and guarded admittance tie on success and avoid the damage proxy. Direct insertion and the no-retract ablation exhibit much larger native peak forces and damage. With only 20 paired seeds per difficulty, these are descriptive cross-physics checks rather than replacements for the primary tests.

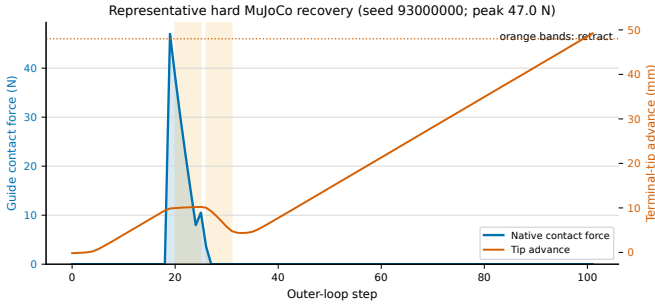


Fig. 5. Representative successful hard-condition native-contact trace. A 47.0-N edge contact triggers two retract intervals, after which the terminal realigns and advances to the seating depth. This trace is regenerated from the hashed MuJoCo report.

another synthetic percentage. It is a preregistered, force-calibrated hardware study with real connector latching and cable reaction loads.

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